Block practice · More homeostasis: skin, kidney and liver

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Identify skin structures and state what each does

Evaluate kidney transplants and dialysis as treatments

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Practice workbook · Identify skin structures and state what each does

Naming skin layers and parts from a diagram or context and giving their roles: protection, sensing, melanin, fat insulation, barrier to infection.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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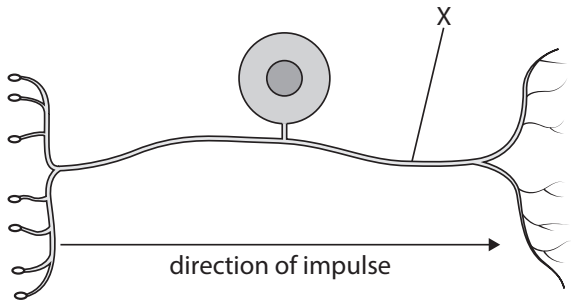
Type ID: HBI-SKIN-STRUCTURE-FUNCTION-RECALL

1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

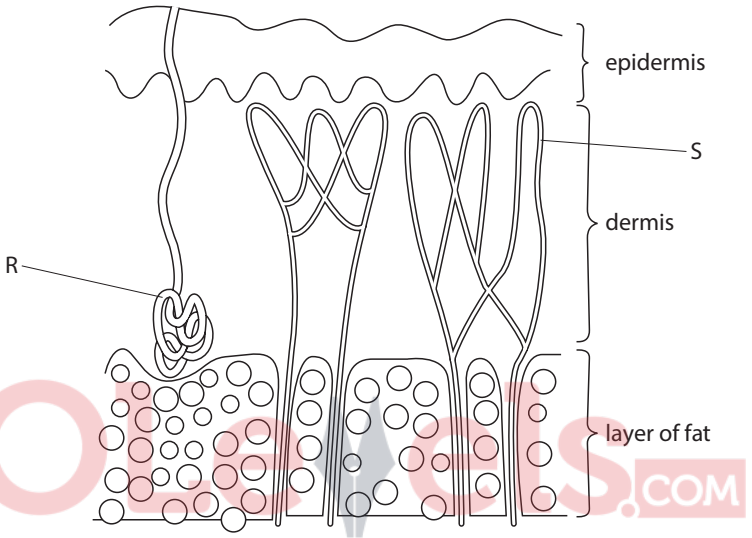
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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R

.....

part S

.....

(ii) Describe how part R helps to cool the body.

(2)

.....

.....

.....

.....

1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

(iii) Describe the role of the layer of fat in maintaining body temperature.

(2)

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1. Question 1 · part b.iii · 2 marks

June 2024 Paper 1

Worked-example retry · Attempt part b.iii only.

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(c) A student investigates the effect of temperature on receptors in the skin.

This is their method.

- place one hand in hot water
- place the other hand in ice cold water
- after 2 minutes take both hands out of the water
- place both hands in water at room temperature

(i) State one possible hazard with their method.

(1)

(ii) Describe how to reduce the risk of this hazard.

(2)

(iii) When the student placed both hands in the water at room temperature the water felt hot to one hand and ice cold to the other.

Name the structure in the skin that detects temperature.

(1)

(Total for Question 1 = 12 marks)

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1. Question 1 · part b.iii · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

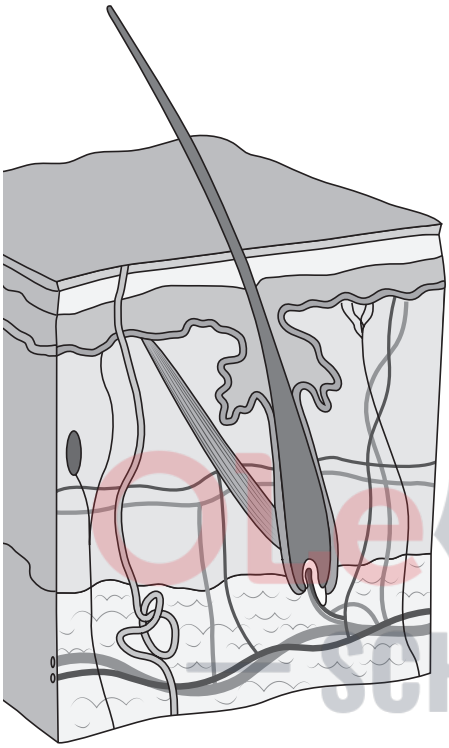
2. Question 5 · part a · 3 marks

January 2020 Paper 2

Further practice · Attempt part a only.

5 Burns can be first degree, second degree or third degree depending on how many layers of the skin are damaged.

The diagram shows the structure of the skin and the layers damaged by each type of burn.



layer 1	first degree burn
layers 1 and 2	second degree burn
layers 1, 2 and 3	third degree burn

(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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2. Question 5 · part a · 3 marks

January 2020 Paper 2

Further practice · Attempt part a only.

(b) Which two skin structures are damaged by a second degree burn?

(1)

- ☒ **A** blood vessels and sebaceous glands
- ☒ **B** fat tissue and blood vessels
- ☒ **C** hair follicles and sweat glands
- ☒ **D** sebaceous glands and fat tissue

(c) Explain why a person with a second or third degree burn has a high risk of infection.

(2)

(d) Blood clotting helps to prevent infection.

Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

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2. Question 5 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

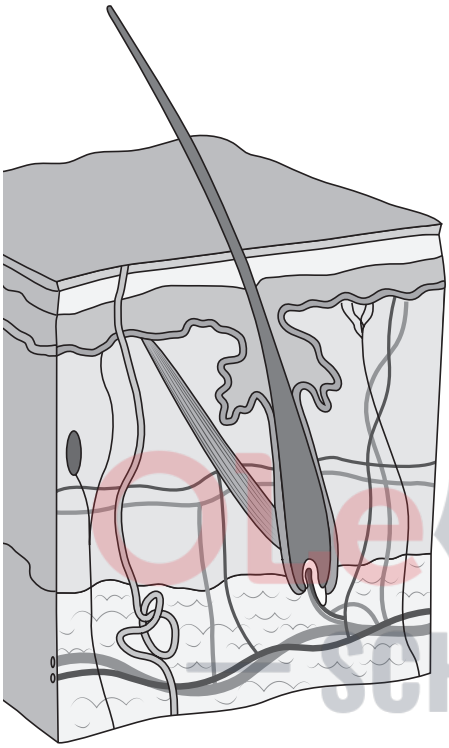
3. Question 5 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

5 Burns can be first degree, second degree or third degree depending on how many layers of the skin are damaged.

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(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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3. Question 5 · part b · 1 mark

January 2020 Paper 2

Further practice · Attempt part b only.

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(b) Which two skin structures are damaged by a second degree burn?

(1)

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(2)

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Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

3. Question 5 · part b · 1 mark · Working and answer

Working and answer

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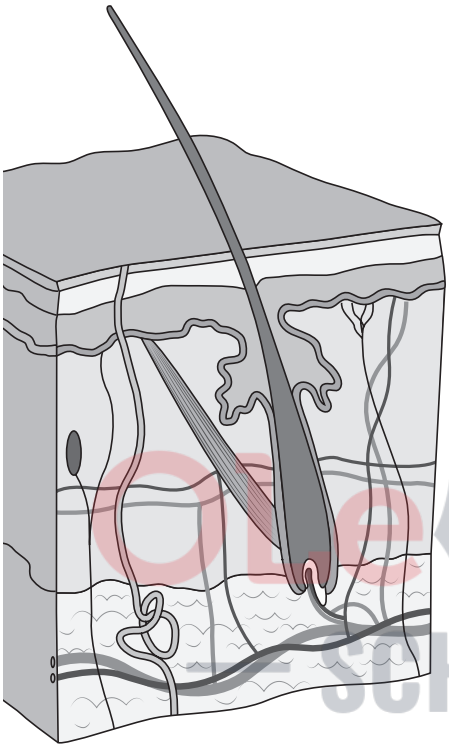
4. Question 5 · part c · 2 marks

January 2020 Paper 2

Further practice · Attempt part c only.

5 Burns can be first degree, second degree or third degree depending on how many layers of the skin are damaged.

The diagram shows the structure of the skin and the layers damaged by each type of burn.



layer 1	first degree burn
layers 1 and 2	second degree burn
layers 1, 2 and 3	third degree burn

(a) Name the three layers of skin that are damaged by a third degree burn.

(3)

layer 1 .....

layer 2 .....

layer 3 .....

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4. Question 5 · part c · 2 marks

January 2020 Paper 2

Further practice · Attempt part c only.

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(b) Which two skin structures are damaged by a second degree burn?

(1)

- ☒ **A** blood vessels and sebaceous glands
- ☒ **B** fat tissue and blood vessels
- ☒ **C** hair follicles and sweat glands
- ☒ **D** sebaceous glands and fat tissue

(c) Explain why a person with a second or third degree burn has a high risk of infection.

(2)

(d) Blood clotting helps to prevent infection.

Describe the process of blood clotting.

(4)

(Total for Question 5 = 10 marks)

4. Question 5 · part c · 2 marks · Working and answer

Working and answer

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5. Question 8 · part a.i · 2 marks

June 2021 Paper 2

Further practice · Attempt part a.i only.

8 Skin and hair contain a pigment called melanin.

(a) (i) Describe the function of the pigment melanin in the skin.

(2)

.....

.....

.....

.....

(ii) State one other place, apart from hair and skin, where pigment is found in the body.

(1)

.....

(b) A condition known as white forelock is caused by a dominant allele H.
People with this condition have a white patch in their hair.

The condition causes a deficiency of the skin pigment melanin.

A woman who is heterozygous for white forelock produces children with a father who does not have the condition.

(i) Draw a genetic diagram to show the possible offspring from this cross.

The condition is not sex-linked.

(4)

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P 6 6 4 3 3 A 0 2 0 2 4

5. Question 8 · part a.i · 2 marks · Working and answer

Working and answer

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6. Question 3 · part a · 2 marks

January 2022 Paper 2

Further practice · Attempt part a only.

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3 The skin is important in the control of body temperature.

(a) State two other functions of the skin.

(2)

1 .....

.....

2 .....

.....

(b) As a result of vigorous exercise, the colour of the skin changes.

Explain why there is this change in skin colour.

(4)

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6. Question 3 · part a · 2 marks · Working and answer

Working and answer

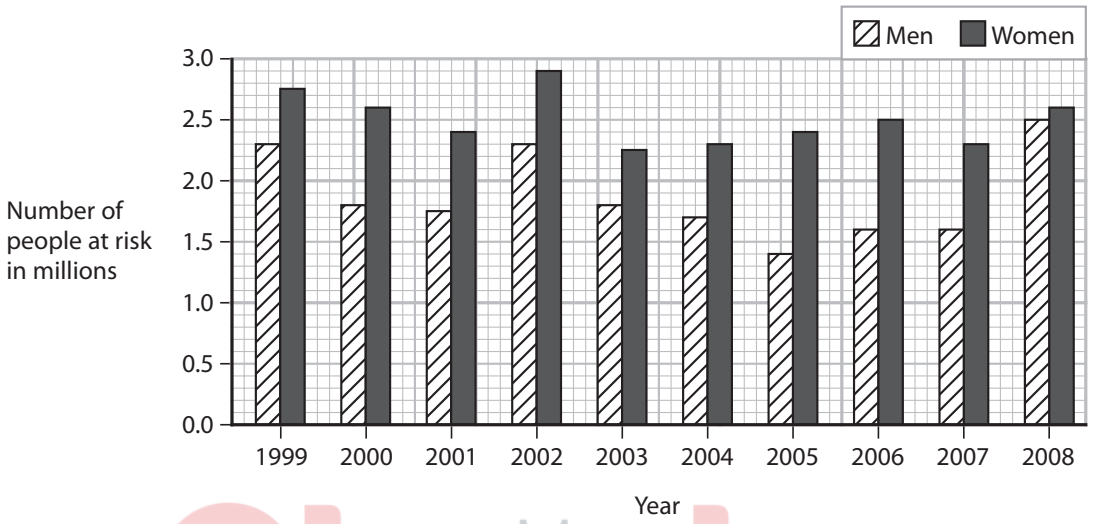
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7. Question 2 · part d · 2 marks

June 2022 Paper 2

Further practice · Attempt part d only.

2 The graph shows the number of working-age people at risk of developing a mental illness.



(a) (i) Determine the year when the greatest number of women were at risk of developing a mental illness.

(1)

.....

(ii) Determine the year when there was the greatest difference between the number of males and the number of females at risk of developing a mental illness.

(1)

.....

(iii) Determine the ratio of females at risk of developing a mental illness in 1999 compared with 2008.

Give your answer in the form n : 1

(2)

ratio = ...

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P 7 0 9 7 4 A 0 4 2 0

7. Question 2 · part d · 2 marks

June 2022 Paper 2

Further practice · Attempt part d only.

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(iv) Suggest one possible reason why the number of females at risk of developing a mental illness increased from 2007 to 2008.

(1)

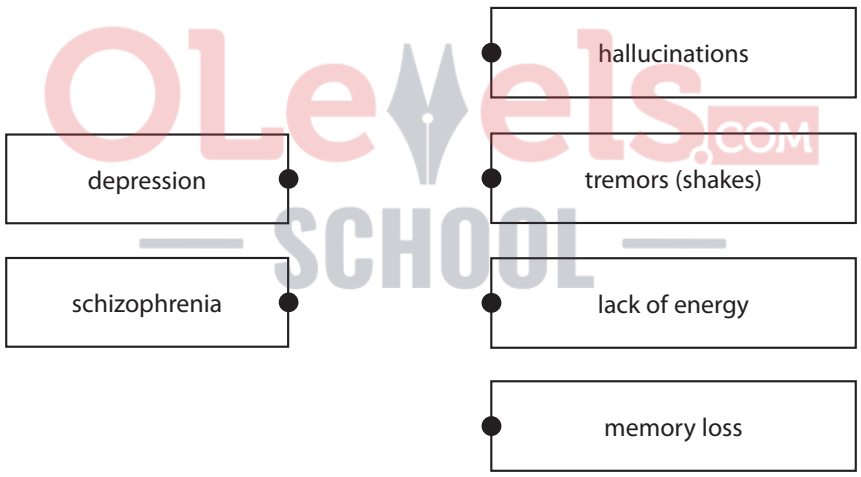
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(b) The boxes list two mental illnesses and some symptoms.

Draw one straight line from each mental illness to a symptom of the illness.

(2)



(c) Which combination of drugs is most likely to affect mental health?

(1)

- ☒ **A** alcohol and heroin
- ☒ **B** antibiotics and alcohol
- ☒ **C** cannabis and paracetamol
- ☒ **D** paracetamol and antibiotics

(d) People who use the drug cocaine often have a raised body temperature.

Name two body structures that help to control body temperature.

(2)

1 .....

2 .....

(Total for Question 2 = 10 marks)

7. Question 2 · part d · 2 marks · Working and answer

Working and answer

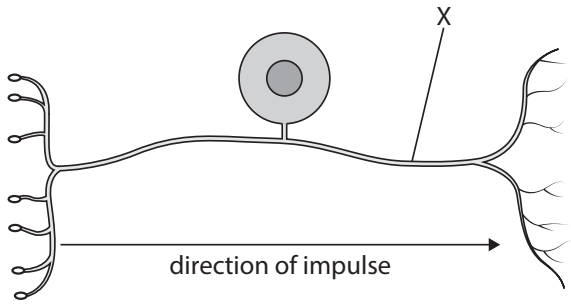
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8. Question 1 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

1 (a) The diagram shows a cell found in the human nervous system.



(Source: © Designua/ Shutterstock)

(i) The cell shown in the diagram transmits electrical impulses from a receptor cell to the central nervous system.

State the name of this type of nerve cell.

(1)

.....

(ii) Name the structure labelled X.

(1)

.....

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8. Question 1 · part b.i · 2 marks

June 2024 Paper 1

Further practice · Attempt part b.i only.

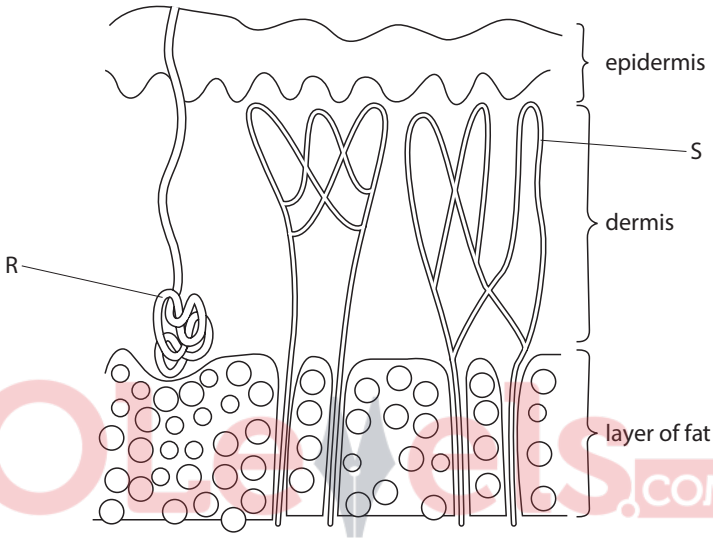
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(b) Structures in the skin respond to information from the nervous system.

The diagram shows a section of the skin.



(Source: adapted from Human Physiology and Health, Author David Wright, Publisher Heinemann page 113)

(i) Name part R and part S.

(2)

part R .....

part S .....

(ii) Describe how part R helps to cool the body.

(2)

.....

.....

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8. Question 1 · part b.i · 2 marks · Working and answer

Working and answer

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Practice workbook · Evaluate kidney transplants and dialysis as treatments

Advantages and disadvantages of kidney transplant and dialysis, and why kidney donation is more common than other organs.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-TRANSPLANT-VS-DIALYSIS

1. Question 8 · part whole · 5 marks

June 2022 Paper 1R

Worked-example retry · Attempt part whole only.

8 The kidneys play an important role in the excretion of toxic waste.

Some people have kidney disease which reduces the effectiveness of the kidneys.

Kidney disease can be treated by dialysis or by a transplant.

Discuss the advantages and disadvantages of each of these treatments.

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(Total for Question 8 = 5 marks)

1. Question 8 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 4 · part b · 5 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

4 The blood is filtered by the kidneys.

The filtrate formed passes through the kidney tubules and is changed into urine.

(a) The table lists the masses of different substances in blood, filtrate and urine passing through the kidneys in one day.

	Mass of substance in g per day		
Substance	**in blood**	**in filtrate**	**in urine**
protein	8000	0	0
glucose	180	180	0
urea	53	53	25
uric acid	9	9	1

Use the information from the table to answer these questions.

(i) Which substance does not pass into the filtrate? **(1)**

- ☒ **A** protein
- ☒ **B** glucose
- ☒ **C** urea
- ☒ **D** uric acid

(ii) The molecules of this substance do not pass into the filtrate.

What is the reason for this? **(1)**

- ☒ **A** the molecules are too large
- ☒ **B** the molecules are reabsorbed
- ☒ **C** the molecules are a waste product
- ☒ **D** the concentration of the molecules in the blood is too low

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2. Question 4 · part b · 5 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

(iii) A person has diabetes but is not receiving treatment.

Explain why the mass of glucose per day in the urine would be different for this person.

(3)

(iv) Calculate the mass of urea in the urine as a percentage of the mass of urea in the filtrate.

(2)

percentage = ... %

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2. Question 4 · part b · 5 marks

June 2019 Paper 1R

Further practice · Attempt part b only.

(b) Kidney failure can be treated by a kidney transplant or by dialysis.

Discuss the advantages and disadvantages of kidney transplants.

(5)

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(Total for Question 4 = 12 marks)

2. Question 4 · part b · 5 marks · Working and answer

Working and answer

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3. Question 7 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

Polycystic kidney disease is caused by a mutation of a gene.

(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

Kidney failure can be treated using dialysis or a transplant.

(i) Describe the benefits of a transplant rather than dialysis as a treatment for kidney failure.

(2)

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3. Question 7 · part b.i · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.i only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

Of these donations, 3233 were kidney donations.

Calculate the percentage of organ donations that are kidney donations.

(2)

percentage = ... %

(iii) The number of kidney donations is much higher than the number of other organ donations such as heart or liver.

Give two reasons for the higher number of kidney donations.

(2)

1 .....

.....

2 .....

.....

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3. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 7 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

7 People with polycystic kidney disease have cysts on their kidneys.

Cysts are sacs of fluid and they affect the functioning of the kidney.

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(a) Explain how a gene mutation can affect the phenotype of an individual.

(3)

(b) Polycystic kidney disease can cause kidney failure.

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(2)

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4. Question 7 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

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(ii) In one year in the United Kingdom there were 4733 organ donations.

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(2)

1 .....

.....

2 .....

.....

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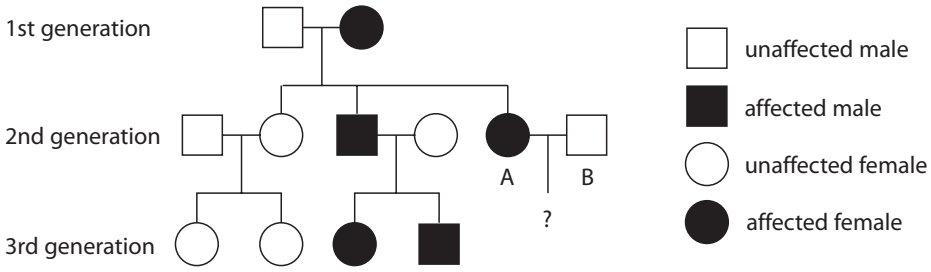
4. Question 7 · part b.iii · 2 marks

January 2023 Paper 1

Further practice · Attempt part b.iii only.

(c) The diagram shows a family pedigree. Some of the family have polycystic kidney disease.

This disease is caused by a dominant allele that is not sex-linked.



(i) Determine the ratio of the females shown in the 3rd generation that are affected with polycystic kidney disease.

(1)

ratio = ...

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4. Question 7 · part b.iii · 2 marks · Working and answer

Working and answer

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5. Question 3 · part c · 2 marks

June 2024 Paper 2R

Further practice · Attempt part c only.

3 (a) Polycystic kidney disease (PKD) is an inherited condition caused by a gene mutation.

PKD affects the function of the kidneys.

State what is meant by a gene mutation.

(1)

.....

.....

(b) PKD occurs when a child inherits a faulty allele from one parent.

One parent has a genotype Hh and the other parent has a genotype hh.

The parents have four children.

Draw a genetic diagram to show the possible genotypes of the four children.

(2)

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(c) Some people with PKD need a kidney transplant.

Give one advantage and one disadvantage of a kidney transplant.

(2)

Advantage

.....

.....

Disadvantage

.....

.....

(Total for Question 3 = 5 marks)

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5. Question 3 · part c · 2 marks · Working and answer

Working and answer

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